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**Defense Commissary Agency (DeCA) — Supply Chain Operations Division**

**Standard Operating Procedure**

**SOP-07: Cold Chain Handling**

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## **1. Purpose**

This Standard Operating Procedure (SOP) establishes the standardized processes, responsibilities, and minimum requirements governing cold chain handling operations across all Defense Commissary Agency (DeCA) distribution centers (DCs) and refrigerated transportation lanes. The primary objective of this SOP is to ensure the maintenance of an unbroken, documented cold chain from the point of receipt of temperature-sensitive product through final delivery to the commissary store level.

This procedure is designed to achieve the following outcomes:

- Standardize cold chain monitoring, response, and documentation procedures across all DeCA DC facilities and transportation partners;
- Ensure uninterrupted refrigeration and temperature integrity for all refrigerated, frozen, and produce product categories entrusted to DeCA custody;
- Ensure full compliance with applicable federal food safety law, including the U.S. Food and Drug Administration (FDA) Food Safety Modernization Act (FSMA) Sanitary Transportation Rule (21 CFR Part 1, Subpart O) and U.S. Department of Agriculture Food Safety and Inspection Service (USDA FSIS) regulations governing the safe handling and transport of meat, poultry, and egg products;

- Protect the food safety, nutritional integrity, and commercial quality of all perishable products distributed through the DeCA supply chain to service members, veterans, retirees, and their families;
- Establish clear Hazard Analysis and Critical Control Points (HACCP) integration requirements, excursion response protocols, escalation procedures, and recordkeeping standards that support both internal audit readiness and regulatory compliance;
- Minimize financial loss attributable to temperature excursions, product spoilage, and cold chain failure events.

This SOP is effective as of 01 August 2026 and supersedes any prior informal cold chain guidance issued at the facility level. All facilities and transportation lanes subject to this SOP shall achieve full compliance no later than 15 August 2026.

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## 2. Scope

This SOP applies to all DeCA distribution center operations and refrigerated transportation lanes involving temperature-sensitive product, including but not limited to refrigerated (perishable), frozen, and produce commodities. The requirements of this document are mandatory for all personnel whose duties bring them into direct or supervisory contact with cold chain assets, product, or monitoring systems.

The following personnel and organizational roles are within scope:

- **DC Cold Storage Operators:** All personnel assigned to walk-in coolers, blast freezers, produce coolers, and refrigerated staging areas at any DeCA distribution center;
- **Receiving Inspectors:** Personnel responsible for receiving inbound shipments and conducting temperature verification at the DC dock;
- **Pickers and Packers:** Warehouse associates selecting and packing perishable and frozen product for outbound orders;

- **Transportation Coordinators:** DeCA staff responsible for scheduling, routing, and monitoring refrigerated carrier activity on DeCA lanes;
- **Refrigeration Technicians:** Certified technicians responsible for maintenance, calibration, monitoring, and first-response to refrigeration and cold chain system alerts;
- **Carriers and Carrier Drivers:** All third-party logistics providers and contract carriers operating refrigerated (reefer) vehicles on DeCA transportation lanes;
- **DC Supervisors:** Shift supervisors and operations managers at each DC facility;
- **Food Safety Officers:** DeCA-designated food safety personnel responsible for HACCP oversight, excursion disposition decisions, and regulatory liaison.

This SOP does not apply to ambient (non-refrigerated) dry grocery product handling, which is addressed under separate DeCA facility operating procedures. This SOP also does not govern store-level refrigeration operations at individual commissary locations, which are subject to commissary-specific food safety plans.

#### ① **SCOPE NOTE**

Any contracted third-party logistics (3PL) provider, transportation carrier, or refrigeration maintenance contractor performing work within DeCA DC facilities or on DeCA transportation lanes is contractually required to comply with the provisions of this SOP as a condition of their contract. Failure to comply constitutes grounds for corrective action, suspension, or contract termination in accordance with the applicable contract terms.

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### 3. References

The following regulatory, statutory, and organizational documents are incorporated by reference and govern the requirements established in this SOP. Personnel responsible for cold chain operations shall maintain familiarity with applicable provisions of each reference as they relate to their assigned duties.

| Reference ID              | Document Title / Description                                                                                    | Issuing Authority                              |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 21 CFR Part 1, Subpart O  | FDA Food Safety Modernization Act (FSMA) — Sanitary Transportation of Human and Animal Food Final Rule          | U.S. Food and Drug Administration (FDA)        |
| FSMA Section 204          | Food Traceability Requirements — Enhanced Traceability for High-Risk Foods                                      | U.S. Food and Drug Administration (FDA)        |
| 9 CFR Part 381 / Part 417 | USDA FSIS Regulations — Poultry Products Inspection / HACCP Systems                                             | USDA Food Safety and Inspection Service (FSIS) |
| DeCA Directive 40-13      | DeCA Food Safety Program — Governing directive for food safety policy across all DeCA operations                | Defense Commissary Agency                      |
| DeCA-SC-SOP-01            | DeCA SOP-01: Receiving & Inspection — Inbound shipment receiving, inspection, and acceptance procedures         | DeCA Supply Chain Operations                   |
| DeCA-SC-SOP-03            | DeCA SOP-03: Order Fulfillment — Pick, pack, and staging procedures for perishable and frozen orders            | DeCA Supply Chain Operations                   |
| DeCA-SC-SOP-04            | DeCA SOP-04: Transportation & Carrier Rate Management — Carrier requirements, lane management, and transit SLAs | DeCA Supply Chain Operations                   |
| NSF/ANSI 7                | Commercial Refrigerators                                                                                        | NSF International / ANSI                       |

| Reference ID            | Document Title / Description                                                                                            | Issuing Authority                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|                         | and Freezers — Performance standards for refrigeration equipment used in food service and storage                       |                                                                           |
| ASHRAE Standard 62.1    | Ventilation and Acceptable Indoor Air Quality — Applicable to refrigerated storage facility ventilation design          | American Society of Heating, Refrigerating and Air-Conditioning Engineers |
| FAR 4.703               | Federal Acquisition Regulation — Contractor Record Retention requirements (3-year minimum for financial impact records) | General Services Administration / DoD                                     |
| DoD Instruction 5400.11 | DoD Privacy Program — Governs safeguarding of FOUO documents                                                            | Office of the Secretary of Defense                                        |

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## 4. Definitions

For the purposes of this Standard Operating Procedure, the following terms and definitions shall apply. Where a term is defined differently in a referenced regulation or directive, the regulatory definition shall govern for matters of regulatory compliance, and this SOP's definition shall govern for operational purposes within DeCA facilities.

| Term              | Definition                                                                                                                                                                                                                                                                                                                                                           |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cold Chain</b> | The unbroken, continuously managed sequence of refrigerated or frozen storage and transport conditions applied to temperature-sensitive food products from the point of manufacturer/origin through receipt, storage, handling, transport, and final delivery. Any break in the cold chain constitutes a temperature excursion event requiring documented corrective |

| Term                         | Definition                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | action.                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Temperature Excursion</b> | Any instance in which the measured temperature of a cold storage asset or refrigerated transport unit exceeds the established acceptable temperature range for that asset, regardless of duration. All excursions must be documented in the Cold Chain Management System (CCMS). Excursions exceeding 30 minutes in duration or 5°F above the maximum acceptable limit require immediate supervisor notification. |
| <b>Set Point</b>             | The target temperature, expressed in degrees Fahrenheit (°F), to which a refrigeration or freezer unit is programmed and controlled. Set points are established by DeCA engineering and are distinct from the acceptable range, which represents the allowable operating variance around the set point.                                                                                                           |
| <b>Reefer Unit</b>           | A refrigerated trailer or container unit equipped with a mechanical refrigeration system, used for the transport of temperature-sensitive products over road, rail, or intermodal transportation lanes. All reefer units operating on DeCA lanes must be equipped with a continuous data logger and pre-cooled to required temperatures prior to loading.                                                         |
| <b>Blast Freezer</b>         | A high-capacity, low-temperature freezer asset designed to rapidly lower product core temperatures to or below 0°F. DeCA blast freezers (Freezer-E1, E2, E3) operate at a set point of 0°F with an acceptable range of -10°F to +5°F and are used for long-term storage of frozen commodities.                                                                                                                    |
| <b>Walk-in Cooler</b>        | A large, walk-in refrigerated storage room maintained for the storage of fresh meat, poultry, seafood, and dairy products. DeCA walk-in coolers (Cooler-A1 through A5 and B1 through B4) operate at a set point of 36°F with an acceptable range of 32°F to 40°F.                                                                                                                                                 |
| <b>Produce Cooler</b>        | A refrigerated storage room specifically                                                                                                                                                                                                                                                                                                                                                                          |



| Term                                | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | designed and maintained for fresh produce commodities, with temperature and humidity controls calibrated to preserve produce freshness and prevent condensation damage. DeCA Produce Cooler (Produce-F1) operates at a set point of 38°F with an acceptable range of 35°F to 45°F.                                                                                                                                                                                                     |
| <b>Data Logger</b>                  | An electronic instrument, either standalone or networked, that continuously records ambient temperature data within a cold storage asset or transport unit at defined intervals (minimum every 30 minutes; configurable to 15 minutes for high-risk assets). All DeCA data loggers must be NIST-traceable and calibrated annually. Logger readings are the primary evidentiary record for cold chain compliance and excursion analysis.                                                |
| <b>HACCP</b>                        | Hazard Analysis and Critical Control Points. A systematic, science-based preventive approach to food safety that identifies physical, chemical, and biological hazards in production and storage processes, and establishes critical control points (CCPs) where control measures can be applied to prevent or eliminate food safety hazards. HACCP plans are legally required for meat and poultry processing and are adopted by DeCA as best practice for all cold chain operations. |
| <b>Critical Control Point (CCP)</b> | A step or procedure in the cold chain handling process at which a control measure must be applied to prevent, eliminate, or reduce a food safety hazard to an acceptable level. DeCA cold chain CCPs include: CCP-1 (Receiving Temperature Check), CCP-2 (Storage Temperature Monitoring), and CCP-3 (Transport Pre-Cool Verification). See Section 13 for full CCP definitions.                                                                                                       |
| <b>Corrective Action</b>            | A documented procedure taken in response to a deviation at a CCP or a temperature excursion event. Corrective actions address immediate                                                                                                                                                                                                                                                                                                                                                |

| Term                                  | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | product safety (hold, evaluate, disposition), identify and eliminate the root cause of the deviation, and restore the system to controlled conditions. All corrective actions are recorded in CCMS and retained per Section 15.                                                                                                                                                                                                                                                    |
| <b>Mean Kinetic Temperature (MKT)</b> | A single calculated temperature that, if maintained over a specified period of time, would produce the same cumulative thermal effect on a product as a series of varying temperature exposures experienced during actual storage or transport. MKT is used by the DeCA Food Safety Officer to evaluate the cumulative thermal impact of a temperature excursion when making product safety disposition decisions. MKT is calculated in accordance with ASTM E1445.                |
| <b>Pre-Cooling</b>                    | The process of bringing a reefer unit's internal temperature down to or below the required loading temperature prior to arrival at the DC dock for loading. Carriers are required to pre-cool reefer units to $\leq 35^{\circ}\text{F}$ (for perishable loads) or $\leq -5^{\circ}\text{F}$ (for frozen loads) before presentation at the DeCA DC dock. Pre-cool temperature is verified and documented by the DC dock supervisor prior to the commencement of loading operations. |

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## 5. Roles and Responsibilities

The following table defines the primary cold chain responsibilities for each organizational role within DeCA DC operations and transportation. These responsibilities are mandatory and non-delegable unless specifically authorized by the DC Director in writing. Individuals in each role shall be trained and certified as required in Section 17 prior to performing the listed duties independently.

| Role                                                     | Primary Cold Chain Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Reporting Authority                 |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Refrigeration Technician</b> (e.g., TECH-08, TECH-12) | <p>Complete daily cold chain checklist (DeCA Form 2210-CC) at 06:00, 12:00, and 18:00 for all cold assets</p> <p>Monitor Cold Chain Dashboard for real-time alerts and alarm conditions</p> <p>Respond to all excursion alarms within 15 minutes</p> <p>Perform routine preventive maintenance on all refrigeration assets</p> <p>Download and review data logger records daily; upload to CCMS</p> <p>Coordinate with certified refrigeration contractors for scheduled PM and emergency repairs</p> <p>Maintain calibration records for all data loggers and temperature instruments</p> <p>Complete root cause analysis (RCA) within 24 hours of confirmed excursion event</p> | DC Supervisor / Food Safety Officer |
| <b>Cold Chain Monitor</b>                                | <p>Continuously monitor the DeCA Cold Chain Dashboard during assigned shift</p> <p>Immediately notify Refrigeration Technician and DC Supervisor upon alarm activation</p> <p>Log all alarm events, notifications, and initial response actions in CCMS</p> <p>Verify alarm acknowledgment and technician response within required 15-minute window</p> <p>Generate daily temperature compliance summary report from CCMS</p>                                                                                                                                                                                                                                                     | DC Supervisor                       |
| <b>Receiving Inspector</b>                               | Verify pre-cool                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | DC Supervisor                       |

| Role                              | Primary Cold Chain Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Reporting Authority                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
|                                   | <p>temperature of all inbound reefer units prior to breaking seals Inspect and record product temperature of all inbound temperature-sensitive shipments using calibrated probe thermometer Reject non-conforming product and initiate Non-Conformance Report (NCR) per SOP-01 and Section 11 of this SOP Document all receiving temperature data in WMS and CCMS Monitor dock staging time; ensure perishables are not held at ambient dock for more than 10 minutes</p>                                                                          |                                                  |
| <b>Picker / Packer</b>            | <p>Retrieve perishable and frozen product from designated cold storage zones only; return to cold storage between pick runs exceeding 10 minutes in ambient areas Report any observed temperature anomalies, equipment malfunctions, or door seal issues immediately to DC Supervisor or Refrigeration Technician Adhere to FIFO (First-In, First-Out) rotation within all cold storage zones Never prop open walk-in cooler or freezer doors for extended periods Comply with all cold storage GMP (Good Manufacturing Practice) requirements</p> | DC Supervisor                                    |
| <b>Transportation Coordinator</b> | <p>Verify carrier compliance with pre-cool requirements prior to authorizing dock</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | DC Supervisor / Supply Chain Operations Division |

| Role                       | Primary Cold Chain Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Reporting Authority                              |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
|                            | <p>assignment Confirm carrier has installed and activated data logger prior to loading Receive and review transit temperature download upon delivery; upload to CCMS Initiate excursion response process for any in-transit temperature deviation identified at delivery Maintain carrier cold chain compliance scorecards; escalate non-compliant carriers per SOP-04 Coordinate with Food Safety Officer for disposition of temperature-compromised inbound loads</p>                                                                                           |                                                  |
| <b>DC Supervisor</b>       | <p>Ensure 100% completion of daily cold chain checklists by assigned Refrigeration Technicians Review CCMS daily temperature compliance summary at start of each shift Authorize product holds in WMS for affected zones upon excursion notification Notify Food Safety Officer immediately upon any excursion exceeding 30 minutes or 5°F above maximum acceptable limit Ensure corrective actions are implemented, verified, and closed within required timeframes Ensure all cold chain personnel maintain required training certifications per Section 17</p> | DC Director / Director, Logistics & Distribution |
| <b>Food Safety Officer</b> | <p>Serve as the primary HACCP authority for DC</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | DC Director / DeCA HQ Food Safety Program        |

| Role                  | Primary Cold Chain Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Reporting Authority                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
|                       | <p>cold chain operations<br/>Evaluate all temperature excursion events involving potential product safety impact using MKT analysis and applicable regulatory standards Make and document final product disposition decisions (release, rework, destroy) for all excursion-affected product Notify FDA and/or USDA FSIS as required by FSMA 204 and applicable regulations Lead quarterly HACCP plan reviews; maintain cold chain HACCP documentation Serve as primary point of contact for FDA/USDA regulatory inspections related to cold chain</p>                          |                                                      |
| <b>Carrier Driver</b> | <p>Pre-cool reefer unit to required temperature prior to DC dock arrival; present pre-cool documentation to dock supervisor Ensure data logger is active and recording prior to loading; do not tamper with logger settings in transit Maintain required set-point temperature throughout transit; report any refrigeration system malfunction to carrier dispatch and DeCA Transportation Coordinator immediately Provide data logger download to receiving DC upon delivery Adhere to all DeCA dock safety and cold chain requirements at loading and delivery locations</p> | Carrier Management / DeCA Transportation Coordinator |

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## 6. Temperature Zones and Standards

All DeCA distribution center cold storage assets are assigned to a designated temperature zone based on the product categories they are authorized to store. Each asset is identified by a unique asset ID, associated with a specific data logger serial number, and subject to the temperature standards defined in the tables below. Asset IDs and logger serial assignments are maintained in the DeCA Asset Register within CCMS and shall be updated whenever equipment is added, replaced, or reconfigured.

Temperature readings recorded by data loggers are the authoritative record for compliance determinations. Manual thermometer readings taken during daily checks (Section 8) serve as a secondary verification and shall be compared against logger readings to detect any instrument drift.

### 6.1 Blast Freezer Zone (Zone E — Long-Term Frozen Storage)

| Asset ID | Zone Name  | Logger Serial | Set Point | Acceptable Range | Early Warning Alarm Threshold | Product Examples                                 |
|----------|------------|---------------|-----------|------------------|-------------------------------|--------------------------------------------------|
| FRZ-E1   | Freezer-E1 | SN-LOG-4431   | 0°F       | -10°F to +5°F    | +3°F (upper); -8°F (lower)    | Frozen Shrimp, Frozen Crab, Frozen Lobster Tails |
| FRZ-E2   | Freezer-E2 | SN-LOG-4432   | 0°F       | -10°F to +5°F    | +3°F (upper); -8°F (lower)    | Frozen Pizza, Frozen Lasagna, Frozen Burritos    |
| FRZ-E3   | Freezer-E3 | SN-LOG-4433   | 0°F       | -10°F to +5°F    | +3°F (upper); -               | Frozen Broccoli,                                 |

| Asset ID | Zone Name | Logger Serial | Set Point | Acceptable Range | Early Warning Alarm Threshold | Product Examples                           |
|----------|-----------|---------------|-----------|------------------|-------------------------------|--------------------------------------------|
|          |           |               |           |                  | 8°F (lower)                   | Frozen Peas, Frozen Corn, Mixed Vegetables |

## 6.2 Walk-in Cooler Zone A — Meat, Poultry, and Seafood

| Asset ID | Zone Name | Logger Serial | Set Point | Acceptable Range | Early Warning Alarm Threshold | Product Examples                                          |
|----------|-----------|---------------|-----------|------------------|-------------------------------|-----------------------------------------------------------|
| REF-A1   | Cooler-A1 | SN-LOG-4421   | 36°F      | 32°F to 40°F     | 38°F (upper); 34°F (lower)    | Fresh Ground Beef, Beef Chuck, Ribeye Steaks              |
| REF-A2   | Cooler-A2 | SN-LOG-4422   | 36°F      | 32°F to 40°F     | 38°F (upper); 34°F (lower)    | Fresh Pork Loin, Pork Ribs, Pork Sausage                  |
| REF-A3   | Cooler-A3 | SN-LOG-4423   | 36°F      | 32°F to 40°F     | 38°F (upper); 34°F (lower)    | Fresh Whole Chicken, Chicken Breasts, Turkey              |
| REF-A4   | Cooler-A4 | SN-LOG-4424   | 36°F      | 32°F to 40°F     | 38°F (upper); 34°F (lower)    | Fresh Atlantic Salmon, Tilapia, Cod Fillets, Shrimp (Raw) |



| Asset ID | Zone Name | Logger Serial | Set Point | Acceptable Range | Early Warning Alarm Threshold | Product Examples                                      |
|----------|-----------|---------------|-----------|------------------|-------------------------------|-------------------------------------------------------|
| REF-A5   | Cooler-A5 | SN-LOG-4425   | 36°F      | 32°F to 40°F     | 38°F (upper); 34°F (lower)    | Processed Deli Meats, Hot Dogs, Bacon, Smoked Sausage |

### 6.3 Walk-in Cooler Zone B — Dairy and Eggs

| Asset ID | Zone Name | Logger Serial | Set Point | Acceptable Range | Early Warning Alarm Threshold | Product Examples                                           |
|----------|-----------|---------------|-----------|------------------|-------------------------------|------------------------------------------------------------|
| REF-B1   | Cooler-B1 | SN-LOG-4426   | 36°F      | 32°F to 40°F     | 38°F (upper); 34°F (lower)    | Fluid Milk (Whole, 2%, Skim), Lactose-Free Milk            |
| REF-B2   | Cooler-B2 | SN-LOG-4427   | 36°F      | 32°F to 40°F     | 38°F (upper); 34°F (lower)    | Shredded Cheese, Block Cheese, Sliced Cheese, Cream Cheese |
| REF-B3   | Cooler-B3 | SN-LOG-4428   | 36°F      | 32°F to 40°F     | 38°F (upper); 34°F (lower)    | Butter, Margarine, Sour Cream, Cottage Cheese              |
| REF-B4   | Cooler-B4 | SN-LOG-4429   | 36°F      | 32°F to 40°F     | 38°F (upper); 34°F (lower)    | Shell Eggs (Grade A, Large), Liquid Eggs, Egg              |

| Asset ID | Zone Name | Logger Serial | Set Point | Acceptable Range | Early Warning Alarm Threshold | Product Examples |
|----------|-----------|---------------|-----------|------------------|-------------------------------|------------------|
|          |           |               |           |                  |                               | Whites           |

## 6.4 Produce Cooler Zone F

| Asset ID | Zone Name  | Logger Serial | Set Point | Acceptable Range | Early Warning Alarm Threshold | Product Examples                                                                     |
|----------|------------|---------------|-----------|------------------|-------------------------------|--------------------------------------------------------------------------------------|
| PRD-F1   | Produce-F1 | SN-LOG-4430   | 38°F      | 35°F to 45°F     | 43°F (upper); 36°F (lower)    | Romaine Lettuce, Spinach, Tomatoes, Broccoli Crowns, Strawberries, Bagged Salad Kits |

## 6.5 Refrigerated Transportation (Reefer Lanes)

| Asset ID | Lane                  | Logger Serial | Load Type               | Required Pre-Cool Temp | Set Point (Transit) | Acceptable Transit Range | Alarm Threshold |
|----------|-----------------------|---------------|-------------------------|------------------------|---------------------|--------------------------|-----------------|
| TRK-001  | LN-001 through LN-005 | SN-LOG-4434   | Perishable (Meat/Dairy) | ≤35°F                  | 36°F                | 32°F to 40°F             | 38°F (upper)    |
| TRK-002  | LN-004 through LN-008 | SN-LOG-4435   | Frozen                  | ≤-5°F                  | 0°F                 | -10°F to +5°F            | +3°F (upper)    |
| TRK-003  | LN-009 through LN-012 | SN-LOG-4436   | Produce                 | ≤40°F                  | 38°F                | 35°F to 45°F             | 43°F (upper)    |

 **CRITICAL: TEMPERATURE ZONE SEPARATION**

Under no circumstances shall refrigerated meat/poultry/seafood product be co-stored with or staged adjacent to dairy, produce, or frozen product unless explicitly authorized by the Food Safety Officer in writing and documented in CCMS. Cross-zone contamination constitutes a food safety violation and shall be reported and corrected immediately.

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## 7. Data Logger Requirements

Data loggers are the primary instrument for cold chain compliance documentation. All loggers used on DeCA cold storage assets and transportation units must meet the standards specified in this section. Non-compliant loggers are grounds for immediate corrective action, and product stored or transported under a non-compliant logger may not be released to store-level distribution until compliance is restored and verified.

### 7.1 Equipment Standards

All cold storage assets must be equipped with a calibrated, NIST-traceable data logger meeting the following minimum requirements:

- NIST-traceable calibration certificate current within the past 12 months; calibration performed annually by an accredited calibration laboratory;
- Temperature measurement accuracy of  $\pm 0.5^{\circ}\text{F}$  or better across the full operating range of the assigned asset;
- Battery backup sufficient to maintain recording for a minimum of 72 hours in the event of facility power loss;
- Tamper-evident seal; any evidence of tampering shall be treated as a potential data integrity violation and reported to the Food Safety Officer;

- Compliant with NSF/ANSI 7 requirements for use in commercial refrigerated environments.

## **7.2 Logging Interval**

The minimum logging interval for all DeCA cold chain assets is one (1) reading per 30 minutes. High-risk assets, as designated by the Food Safety Officer (including assets with a history of recurrent excursions or assets storing HACCP-regulated product categories), shall be configured to log at 15-minute intervals. Current high-risk asset designations are maintained by the Food Safety Officer in CCMS and are reviewed quarterly.

## **7.3 Logger Registration and Asset Association**

Each data logger serial number shall be registered in the DeCA CCMS Asset Register and associated with its assigned cold storage asset or transportation unit. The following information shall be recorded for each logger:

- Logger serial number (SN-LOG-XXXX format);
- Assigned asset ID (e.g., REF-A1, FRZ-E1);
- Date of installation and date of last calibration;
- Calibration laboratory name and certificate number;
- Logging interval setting (15 or 30 minutes);
- Alarm threshold configuration (must match Section 6 standards for assigned asset).

## **7.4 Daily Data Download and Review**

The assigned Refrigeration Technician (TECH-08 or TECH-12 per shift assignment) shall download data from all stationary asset loggers daily during the 06:00 check and upload the data to the CCMS Cold Chain Dashboard by no later than 07:30 daily. Transit logger data shall be downloaded from carrier-provided logs upon delivery and uploaded to CCMS by the Transportation Coordinator within two (2) hours of delivery completion. All downloaded data shall be reviewed for any unresolved excursion events not previously captured by real-time alarms.

## 7.5 Wireless and Automated Logger Transmission

Wireless-enabled loggers installed on stationary cold storage assets (REF-A1 through A5, REF-B1 through B4, FRZ-E1 through E3, PRD-F1) shall transmit temperature readings in real-time to the DeCA Cold Chain Dashboard via the facility's secured wireless network. Dashboard monitoring is the primary means of excursion detection. Automated alarms shall be configured at the thresholds specified in Section 6 and shall generate both an audible alert in the monitoring station and an automated notification to the on-duty Refrigeration Technician and Cold Chain Monitor via the CCMS notification system.

## 7.6 Logger Serial Number Reference Table

The following table constitutes the official logger serial-to-asset cross-reference for this facility as of the effective date of this SOP. Any changes to logger assignments must be updated in CCMS within 24 hours of the change and reported to the Food Safety Officer.

| Logger Serial Number | Assigned Asset ID | Zone / Asset Name        | Zone Type      | Logging Interval   | Last Calibration Date |
|----------------------|-------------------|--------------------------|----------------|--------------------|-----------------------|
| SN-LOG-4421          | REF-A1            | Cooler-A1 (Meat)         | Walk-in Cooler | 15 min (high-risk) | 15 Jan 2026           |
| SN-LOG-4422          | REF-A2            | Cooler-A2 (Pork)         | Walk-in Cooler | 30 min             | 15 Jan 2026           |
| SN-LOG-4423          | REF-A3            | Cooler-A3 (Poultry)      | Walk-in Cooler | 15 min (high-risk) | 15 Jan 2026           |
| SN-LOG-4424          | REF-A4            | Cooler-A4 (Seafood)      | Walk-in Cooler | 15 min (high-risk) | 15 Jan 2026           |
| SN-LOG-4425          | REF-A5            | Cooler-A5 (Deli)         | Walk-in Cooler | 30 min             | 15 Jan 2026           |
| SN-LOG-4426          | REF-B1            | Cooler-B1 (Milk)         | Walk-in Cooler | 30 min             | 15 Jan 2026           |
| SN-LOG-4427          | REF-B2            | Cooler-B2 (Cheese)       | Walk-in Cooler | 30 min             | 15 Jan 2026           |
| SN-LOG-4428          | REF-B3            | Cooler-B3 (Butter/Cream) | Walk-in Cooler | 30 min             | 15 Jan 2026           |
| SN-LOG-4429          | REF-B4            | Cooler-B4 (Eggs)         | Walk-in Cooler | 30 min             | 15 Jan 2026           |

| Logger Serial Number | Assigned Asset ID | Zone / Asset Name             | Zone Type              | Logging Interval    | Last Calibration Date |
|----------------------|-------------------|-------------------------------|------------------------|---------------------|-----------------------|
| SN-LOG-4430          | PRD-F1            | Produce-F1                    | Produce Cooler         | 30 min              | 15 Jan 2026           |
| SN-LOG-4431          | FRZ-E1            | Freezer-E1                    | Blast Freezer          | 30 min              | 15 Jan 2026           |
| SN-LOG-4432          | FRZ-E2            | Freezer-E2                    | Blast Freezer          | 30 min              | 15 Jan 2026           |
| SN-LOG-4433          | FRZ-E3            | Freezer-E3                    | Blast Freezer          | 30 min              | 15 Jan 2026           |
| SN-LOG-4434          | TRK-001           | Reefer Truck 001 (Perishable) | Refrigerated Transport | 30 min (continuous) | 01 Feb 2026           |
| SN-LOG-4435          | TRK-002           | Reefer Truck 002 (Frozen)     | Refrigerated Transport | 30 min (continuous) | 01 Feb 2026           |
| SN-LOG-4436          | TRK-003           | Reefer Truck 003 (Produce)    | Refrigerated Transport | 30 min (continuous) | 01 Feb 2026           |

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## 8. Monitoring and Daily Check Procedure

The daily cold chain check is a mandatory operational procedure performed three times per day across all cold storage assets. This procedure supplements the automated real-time monitoring provided by the Cold Chain Dashboard and provides an independent, manual verification that all assets are functioning within required temperature parameters and that all physical components (doors, gaskets, coils, alarms) are in serviceable condition.

### 8.1 Check Schedule and Responsible Personnel

The assigned Refrigeration Technician (TECH-08 for day shift; TECH-12 for swing/evening shift) shall complete the DeCA Form 2210-CC Cold Chain Daily Checklist for all cold storage assets at the following times each operational day:

- **Check 1:** 06:00 local time — Performed by on-duty technician at shift start; includes data logger download;

- **Check 2:** 12:00 local time (Noon) — Midday verification; particular attention to assets with any morning alerts;
- **Check 3:** 18:00 local time — End-of-day verification; confirm all assets are secure for overnight operations.

Checks shall not be skipped, delegated to non-technician personnel, or completed outside a 30-minute window of the scheduled time without written authorization from the DC Supervisor. Missed checks constitute a compliance deficiency and shall be documented in CCMS with an explanation.

## 8.2 Checklist Items — DeCA Form 2210-CC

For each cold storage asset, the Refrigeration Technician shall inspect and record the following items on DeCA Form 2210-CC:

1. **Ambient Temperature:** Recorded from the asset's primary digital display and confirmed against the data logger reading. Variance greater than  $\pm 2^{\circ}\text{F}$  between display and logger reading requires immediate investigation;
2. **Set-Point Verification:** Confirm the asset's controller is programmed to the correct set point per Section 6; document any unauthorized set-point changes;
3. **Door Gasket Inspection:** Visually inspect all door gaskets and seals for tears, gaps, ice buildup, or deformation; note any deficiencies requiring maintenance;
4. **Evaporator Coil Condition:** Inspect coils for excessive ice buildup (indicating possible defrost cycle malfunction), debris, or airflow obstruction;
5. **Alarm System Test:** Verify that the high- and low-temperature alarms are active and set to the correct thresholds per Section 6; document test result as PASS or FAIL;
6. **Door Closure Verification:** Confirm all doors are fully closed, latched, and sealed; door strip curtains are in place where applicable;
7. **Data Logger Status:** Confirm logger is powered, recording, and transmitting to dashboard (for wireless loggers); note battery status.

All check items shall be recorded as PASS, FAIL, or N/A on DeCA Form 2210-CC. Any FAIL result requires a corrective action note in the same form entry and initiation of a CCMS maintenance work order.

### 8.3 Documentation in CCMS

All completed Form 2210-CC entries shall be entered into the Cold Chain Management System (CCMS) by the completing technician within 60 minutes of the physical check. CCMS entries shall include the technician's employee ID, asset ID, check time, all recorded values, pass/fail results, and any corrective action notes. Paper copies of Form 2210-CC shall be retained in the facility's Cold Chain binder (organized by asset and date) as backup and submitted to the DC Supervisor for weekly review.

### 8.4 Representative Daily Check Examples (01 August 2026)

The following table provides representative daily check results as recorded by TECH-08 during the 06:00 check on 01 August 2026, the effective date of this SOP, for assets REF-A1 (Cooler-A1) and FRZ-E1 (Freezer-E1):

| Field                         | REF-A1 (Cooler-A1) —<br>06:00 Check    | FRZ-E1 (Freezer-E1) —<br>06:00 Check                      |
|-------------------------------|----------------------------------------|-----------------------------------------------------------|
| Check Date / Time             | 01 Aug 2026 / 06:04                    | 01 Aug 2026 / 06:12                                       |
| Technician ID                 | TECH-08                                | TECH-08                                                   |
| Asset ID / Logger SN          | REF-A1 / SN-LOG-4421                   | FRZ-E1 / SN-LOG-4431                                      |
| Digital Display Temp          | 35.8°F                                 | -1.2°F                                                    |
| Logger Reading                | 35.6°F                                 | -1.0°F                                                    |
| Variance (Display vs. Logger) | 0.2°F — Within tolerance               | 0.2°F — Within tolerance                                  |
| Set Point Verified            | PASS (36°F confirmed)                  | PASS (0°F confirmed)                                      |
| Door Gasket Inspection        | PASS — No deficiencies noted           | PASS — No deficiencies noted                              |
| Evaporator Coil Condition     | PASS — Clear, no ice buildup           | PASS — Normal frost accumulation; defrost scheduled 08:00 |
| Alarm System Test             | PASS — High alarm at 38°F; Low at 34°F | PASS — High alarm at +3°F; Low at -8°F                    |



| Field                | REF-A1 (Cooler-A1) —<br>06:00 Check         | FRZ-E1 (Freezer-E1) —<br>06:00 Check        |
|----------------------|---------------------------------------------|---------------------------------------------|
| Door Closure         | PASS — All doors secured                    | PASS — All doors secured                    |
| Data Logger Status   | PASS — Transmitting to dashboard            | PASS — Transmitting to dashboard            |
| Overall Result       | <b>PASS — No corrective action required</b> | <b>PASS — No corrective action required</b> |
| CCMS Entry Timestamp | 01 Aug 2026 / 06:58                         | 01 Aug 2026 / 07:02                         |

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## 9. Temperature Excursion Response Procedure

A temperature excursion is any condition in which the measured temperature of a cold storage asset or refrigerated transport unit exceeds or falls below the acceptable range defined in Section 6 of this SOP. All excursions, regardless of duration or cause, are treated as potential food safety events and must be responded to, documented, and closed through the corrective action process. The following procedure applies to all excursion events across all DeCA cold chain assets.

### 9.1 Alarm Detection and Initial Notification

An excursion alarm is triggered automatically by the Cold Chain Dashboard when a data logger reading meets or exceeds the early warning alarm threshold defined for the asset in Section 6. The alarm generates the following simultaneous notifications:

- Audible alarm at the Cold Chain Monitor station;
- Push notification to the on-duty Refrigeration Technician via CCMS mobile alert;
- Automated email to the DC Supervisor and Food Safety Officer (if excursion persists more than 15 minutes);
- Automatic event entry created in CCMS with timestamp, asset ID, and temperature reading.

## 9.2 Technician Response (Within 15 Minutes)

The on-duty Refrigeration Technician (TECH-08 or TECH-12) shall physically present at the affected asset within **15 minutes** of alarm activation to assess the situation. The technician shall:

8. Verify the alarm reading using a calibrated handheld thermometer independent of the data logger;
9. Assess the current temperature trend (rising, stable, or declining);
10. Identify the probable root cause from the following common causes:  
compressor failure or lockout, door left open or improperly sealed, defrost cycle malfunction, condenser fan failure, refrigerant leak, or loss of facility electrical power to the unit;
11. Take immediate corrective action if within technician's technical authority (e.g., close door, reset defrost cycle, restart compressor);
12. Log initial response in CCMS within 15 minutes of physical assessment, documenting arrival time, observed temperature, probable cause, and initial action taken.

## 9.3 Cause Assessment and Documentation

The technician shall document the probable root cause of the excursion in CCMS at the time of initial response. The following information must be captured:

- Asset ID and logger serial number;
- Excursion start time (as recorded by data logger);
- Peak temperature recorded during excursion;
- Duration of excursion from start time to return to acceptable range;
- Probable root cause (selected from CCMS dropdown or entered as free text);
- Initial corrective action taken by technician.

## 9.4 Supervisor and Food Safety Officer Escalation

The DC Supervisor shall be notified by the Refrigeration Technician or Cold Chain Monitor **immediately** if any of the following escalation criteria are met:

- The excursion has persisted for more than 30 minutes from alarm activation;
- The peak recorded temperature exceeds the maximum acceptable limit by more than 5°F (e.g., above 45°F for a perishable cooler; above +10°F for a blast freezer);
- The technician is unable to identify the root cause or restore the asset to normal operation within 30 minutes;
- Any excursion affecting a HACCP-regulated product category (meat, poultry, seafood).

Upon escalation, the DC Supervisor shall immediately authorize a product hold in the WMS for all product located in the affected zone and notify the Food Safety Officer.

## 9.5 Product Hold and WMS Action

Upon receiving notification of an escalated excursion from the DC Supervisor, the Food Safety Officer shall direct that a **WMS product hold** be placed on all product in the affected cold storage zone. The hold prevents any affected product from being picked, packed, shipped, or released until the Food Safety Officer has completed a safety evaluation and issued a disposition decision. The hold is documented in CCMS with the hold authorization timestamp and the Food Safety Officer's employee ID.

## 9.6 Mean Kinetic Temperature (MKT) Analysis and Disposition

The Food Safety Officer shall evaluate the safety of held product using Mean Kinetic Temperature (MKT) analysis calculated from the data logger records of the excursion event. The evaluation shall also consider the product category, estimated remaining shelf life, and applicable regulatory standards (FDA, USDA FSIS). The disposition decision shall be documented in CCMS and shall fall into one of the following categories:

- **Release:** MKT analysis confirms product remains within safe temperature parameters; product hold is lifted; product is cleared for normal distribution;

- **Rework / Reprocess:** Product quality may be affected but food safety is not compromised; product is flagged for rework (e.g., reduced price, accelerated distribution) as authorized by the Food Safety Officer;
- **Destroy / Condemn:** MKT analysis or regulatory standards indicate product cannot be certified as safe for human consumption; product is condemned, removed from inventory, and destroyed per DeCA waste and loss procedures. See Section 14 for full escalation and FDA/USDA notification requirements.

## 9.7 Root Cause Analysis and Corrective Action

For all escalated excursion events, the assigned Refrigeration Technician and DC Supervisor shall complete a documented **Root Cause Analysis (RCA)** within **24 hours** of the excursion event. The RCA shall identify the specific root cause (not merely the symptom), identify contributing factors, and specify the corrective action implemented to prevent recurrence. The RCA and corrective action are entered in CCMS and reviewed by the Food Safety Officer before the event is closed. The DC Supervisor shall verify the corrective action is effective within 72 hours of implementation.

## 9.8 Documented Excursion Examples (July 2026)

The following table presents three confirmed temperature excursion events from the DeCA cold chain temperature log for July 2026, documented as representative examples for training and reference purposes:

| Event #        | Asset ID / Lane                               | Date        | Peak Temp Recorded                            | Duration   | Root Cause                                                       | Technician | Disposition / Outcome                                                     |
|----------------|-----------------------------------------------|-------------|-----------------------------------------------|------------|------------------------------------------------------------------|------------|---------------------------------------------------------------------------|
| EXC-2026-07-01 | REF-A2 (Cooler-A2 — Pork) Logger: SN-LOG-4422 | 05 Jul 2026 | 44°F (Max Acceptable: 40°F; Exceedance: +4°F) | 45 minutes | Compressor alarm — low refrigerant pressure ; compressor entered | TECH-08    | Product hold placed. MKT analysis completed. 45-minute excursion at 44°F: |

| Event #        | Asset ID / Lane                              | Date        | Peak Temp Recorded                                    | Duration   | Root Cause                                                                                                                        | Technician                     | Disposition / Outcome                                                                                                                                                                                                         |
|----------------|----------------------------------------------|-------------|-------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |                                              |             |                                                       |            | safety lockout mode. Technician reset lockout and recharged refrigerant. Certified contract or called for same-day PM inspection. |                                | cumulative thermal load within safe limits per USDA FSIS guidance. Product <b>Released</b> after Food Safety Officer approval. RCA completed; refrigerant leak repaired 05 Jul 2026. Preventive maintenance schedule updated. |
| EXC-2026-07-02 | TRK-001 / Lane LN-003<br>Logger: SN-LOG-4434 | 08 Jul 2026 | 43°F (Max Acceptable Transit: 40°F; Exceedance: +3°F) | 30 minutes | Door seal failure on reefer unit — rear door weather stripping separating                                                         | TECH-12 (reviewed at delivery) | Transit temperature download reviewed at delivery DC. Product hold placed                                                                                                                                                     |

| Event # | Asset ID / Lane | Date | Peak Temp Recorded | Duration | Root Cause                                                                                                                   | Technician | Disposition / Outcome                                                                                                                                                                                                                                                                                                                       |
|---------|-----------------|------|--------------------|----------|------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                 |      |                    |          | ed, allowing warm air infiltration. Driver did not detect until mid-route. Reefer unit repaired upon return to carrier yard. |            | on perishable load (Meat/Dairy). MKT analysis : 30-minute excursion at 43°F within acceptable safety margin. Product <b>Released</b> with expedited store delivery requirement. Carrier notified of violation ; corrective action report required per SOP-04. Door seal replaced by carrier. Non-conformance logged against carrier record. |

| Event #        | Asset ID / Lane                              | Date        | Peak Temp Recorded                                              | Duration   | Root Cause                                                                                                                                                                                                                                                        | Technician                      | Disposition / Outcome                                                                                                                                                                                                                                                                                                             |
|----------------|----------------------------------------------|-------------|-----------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EXC-2026-07-03 | TRK-002 / Lane LN-005<br>Logger: SN-LOG-4435 | 14 Jul 2026 | +8°F (Max Acceptable Transit: +5°F; Exceedance: +3°F above max) | 60 minutes | Refrigeration unit (reefer engine) failure en route — compressor belt failure caused total loss of refrigeration for approximately 60 minutes before driver detected alarm and contacted dispatch . Emergency reefer service dispatched; unit repaired roadside . | TECH-12 (reviewed at delivery ) | Product hold placed immediately upon delivery temperature download review. MKT analysis : 60-minute excursion at +8°F on frozen load determined to constitute unacceptable cumulative thermal load. Food Safety Officer and DC Director notified within 1 hour per Section 14. <b>Load quarantined;</b> entire frozen load condem |

| Event # | Asset ID / Lane | Date | Peak Temp Recorded | Duration | Root Cause | Technician | Disposition / Outcome                                                                                                                                                                                                                                                                                                 |
|---------|-----------------|------|--------------------|----------|------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                 |      |                    |          |            |            | ned and destroyed. Estimated loss documented in CCMS and financial system. Carrier placed on corrective action hold per SOP-04. RCA completed within 24 hours; root cause: deferred preventive maintenance on reefer belt. Carrier required to provide PM compliance documentation before resuming DeCA frozen lanes. |



### **REGULATORY REMINDER**

Any temperature excursion resulting in product condemnation or destruction where the food safety event may affect products distributed to commissary stores must be evaluated against FDA FSMA Section 204 traceability notification requirements. The Food Safety Officer shall make this determination within 2 hours of the disposition decision. When in doubt, notify.

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## **10. Refrigerated Transportation Requirements**

All carriers operating refrigerated transportation lanes on behalf of the Defense Commissary Agency must comply with the following minimum requirements as a condition of their carrier agreement and in accordance with DeCA-SC-SOP-04 and the FDA FSMA Sanitary Transportation Rule (21 CFR Part 1, Subpart O). Non-compliance with any of the requirements in this section constitutes a carrier compliance violation and shall be reported to the DeCA Transportation Coordinator for action under the applicable carrier contract.

### **10.1 Pre-Cool Requirements**

Carrier must pre-cool the reefer unit to the required temperature **prior to arrival at the DeCA DC dock**. Pre-cool temperature requirements are as follows:

- **Perishable loads (meat, dairy, produce):** Reefer unit interior temperature must be at or below **35°F** before dock arrival;
- **Frozen loads:** Reefer unit interior temperature must be at or below **-5°F** before dock arrival.

Carriers shall present pre-cool temperature documentation (printout from reefer unit temperature controller or data logger) to the DC dock supervisor upon arrival.

Pre-cool documentation shall include the asset ID/unit number, pre-cool temperature achieved, and the timestamp of the reading.

## **10.2 Pre-Cool Verification by DC Dock Supervisor**

The DC dock supervisor (or designated receiving inspector) shall verify the reefer unit's pre-cool temperature **before authorizing loading to commence**.

Verification shall be performed using the following process:

13. Review carrier-provided pre-cool documentation;
14. Independently verify interior reefer temperature using a calibrated probe or non-contact thermometer at the dock door before the trailer is positioned and loaded;
15. Record the verified pre-cool temperature, the reefer unit ID, carrier name, date/time, and dock supervisor employee ID on the Loading Authorization Log (DeCA Form 2215-LA);
16. If the reefer unit has not achieved the required pre-cool temperature, loading shall not commence. The carrier shall be directed to continue pre-cooling; if the carrier cannot achieve pre-cool within 30 minutes of scheduled loading time, the load shall be rescheduled and the deviation reported to the Transportation Coordinator.

## **10.3 Perishable Staging Time Limit**

Perishable and frozen products may not be staged at an ambient-temperature dock area for more than **10 minutes** from the time product exits the cold storage asset to the time loading into the pre-cooled reefer unit commences. If the 10-minute limit cannot be met due to dock delays, the product must be returned to cold storage immediately. The dock supervisor is responsible for enforcing and documenting compliance with this requirement.

## **10.4 Continuous Data Logger Requirement for All Reefer Lanes**

A carrier-provided continuous data logger (compliant with the standards in Section 7.1) is required for every refrigerated reefer lane operated on behalf of DeCA. The logger must:

- Be activated and recording before loading commences (start time confirmed on logger display or printout);
- Record temperature at minimum every 30 minutes throughout the transit;
- Not be tampered with, repositioned, or de-activated during transit by the driver or carrier personnel;
- Remain in the reefer unit until the temperature download is completed at the delivery DC.

## **10.5 Temperature Download at Delivery**

Upon arrival at the delivery DC, the carrier driver shall provide a complete data logger download (printout or electronic file) to the receiving DC's Transportation Coordinator or receiving inspector. The download shall cover the full transit period from loading through delivery. A copy shall be retained by both the delivery DC (uploaded to CCMS) and the carrier. Any excursion identified during the transit temperature review shall trigger the excursion response process per Section 9.

## **10.6 Carrier Responsibility for In-Transit Excursions**

The carrier is fully responsible for all temperature excursions occurring during transit from the time the trailer doors are sealed at the loading DC to the time the doors are broken at the delivery DC, except where an excursion is directly attributable to product that was outside acceptable temperature at the time of loading (in which case the receiving DC shares responsibility). All in-transit excursions shall result in the carrier's corrective action obligation per SOP-04, and may result in financial liability for product loss in accordance with the carrier contract.

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# **11. Receiving Temperature Acceptance Standards**

The following temperature acceptance standards apply to all inbound temperature-sensitive shipments received at DeCA distribution centers. These standards are aligned with DeCA-SC-SOP-01 (Receiving & Inspection), FDA FSMA requirements,

and USDA FSIS regulations. All receiving inspectors must be trained on these standards and must apply them consistently at the time of inbound shipment inspection. Product arriving outside the acceptance temperature range shall be rejected and a Non-Conformance Report (NCR) initiated without exception.

| <b>Product Category</b>            | <b>Acceptance Temperature (Max at Receipt)</b> | <b>Rejection Threshold</b>  | <b>NCR Required?</b> | <b>Action Upon Rejection</b>                                                                                   |
|------------------------------------|------------------------------------------------|-----------------------------|----------------------|----------------------------------------------------------------------------------------------------------------|
| Frozen Product (All types)         | ≤ 0°F                                          | > 0°F (any thawing evident) | Yes — Mandatory      | Reject entire lot; do not commingle with in-stock frozen product; return to carrier; document in WMS and CCMS. |
| Fresh Meat (Beef, Pork, Lamb)      | ≤ 40°F                                         | > 40°F                      | Yes — Mandatory      | Reject lot; initiate NCR; notify Food Safety Officer if temp > 45°F.                                           |
| Fresh Poultry (Chicken, Turkey)    | ≤ 40°F                                         | > 40°F                      | Yes — Mandatory      | Reject lot; initiate NCR; notify Food Safety Officer immediately; USDA FSIS notification may be required.      |
| Fresh Seafood (Fish, Shellfish)    | ≤ 40°F (32°F preferred)                        | > 40°F                      | Yes — Mandatory      | Reject lot; initiate NCR; notify Food Safety Officer if temp > 41°F for any duration.                          |
| Dairy (Fluid Milk, Cheese, Butter) | ≤ 40°F                                         | > 40°F                      | Yes — Mandatory      | Reject lot; initiate NCR. Hold and evaluate per                                                                |

| Product Category              | Acceptance Temperature (Max at Receipt) | Rejection Threshold | NCR Required?            | Action Upon Rejection                                                                                       |
|-------------------------------|-----------------------------------------|---------------------|--------------------------|-------------------------------------------------------------------------------------------------------------|
|                               |                                         |                     |                          | MKT if temp 40–45°F for <30 min.                                                                            |
| Shell Eggs                    | ≤ 40°F (USDA Grade A requirement)       | > 40°F              | Yes — Mandatory          | Reject lot; initiate NCR; notify Food Safety Officer.                                                       |
| Fresh Produce                 | ≤ 45°F                                  | > 45°F              | Yes — Mandatory if >50°F | Hold for Food Safety Officer evaluation if 45–50°F; reject if >50°F or product shows visible deterioration. |
| Processed Deli / RTE Products | ≤ 40°F                                  | > 40°F              | Yes — Mandatory          | Reject lot; initiate NCR; treat as high-risk Ready-to-Eat; notify Food Safety Officer.                      |

### **NCR INITIATION REQUIREMENT**

A Non-Conformance Report (NCR) must be initiated in the DeCA WMS within 30 minutes of any product rejection at the dock. The NCR shall document: vendor/carrier name, product description, lot number and quantity, recorded temperature at rejection, receiving inspector employee ID, and the disposition action taken. NCRs are reviewed weekly by the DC Supervisor and monthly by the Supply Chain Operations Division.

## 12. Maintenance and Calibration Schedule

All cold storage assets and data loggers are subject to a structured preventive maintenance (PM) and calibration program designed to ensure continuous operational readiness, minimize unplanned excursion events, and sustain regulatory compliance. The following table defines the mandatory maintenance and calibration activities, frequency, responsible party, and applicable assets.

| Activity                                                      | Frequency                                                      | Responsible Party                                                            | Applicable Assets                                               | Documentation Required                               |
|---------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|
| Daily Cold Chain Checklist (Form 2210-CC)                     | Daily — 3x per day (06:00, 12:00, 18:00)                       | Refrigeration Technician (TECH-08 / TECH-12)                                 | REF-A1 through A5, REF-B1 through B4, FRZ-E1 through E3, PRD-F1 | DeCA Form 2210-CC; CCMS entry within 60 min of check |
| Data Logger Download and Review                               | Daily (06:00 check)                                            | Refrigeration Technician                                                     | SN-LOG-4421 through SN-LOG-4436                                 | CCMS upload; logger download file retained in CCMS   |
| Door Gasket and Seal Replacement (as needed / minimum annual) | Annual minimum; replace immediately upon any failed inspection | Refrigeration Technician (minor); Certified Refrigeration Contractor (major) | REF-A1 through A5, REF-B1 through B4, FRZ-E1 through E3, PRD-F1 | CCMS maintenance work order; parts record            |
| Evaporator Coil Cleaning                                      | Monthly                                                        | Certified Refrigeration Contractor                                           | REF-A1 through A5, REF-B1 through B4, FRZ-E1 through E3, PRD-F1 | PM completion certificate; CCMS entry                |
| Condenser Cleaning and Inspection                             | Monthly                                                        | Certified Refrigeration Contractor                                           | REF-A1 through A5, REF-B1 through B4, FRZ-E1 through E3, PRD-F1 | PM completion certificate; CCMS entry                |
| Refrigerant Level Check                                       | Monthly                                                        | Certified Refrigeration                                                      | REF-A1 through A5,                                              | Refrigerant log; EPA                                 |

| Activity                                               | Frequency                                       | Responsible Party                                       | Applicable Assets                                               | Documentation Required                                                                              |
|--------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| and Leak Test                                          |                                                 | Contractor (EPA 608 certified)                          | REF-B1 through B4, FRZ-E1 through E3, PRD-F1                    | Section 608 leak inspection record; CCMS entry                                                      |
| Compressor and Electrical Systems Inspection           | Monthly                                         | Certified Refrigeration Contractor                      | REF-A1 through A5, REF-B1 through B4, FRZ-E1 through E3, PRD-F1 | PM completion certificate; CCMS entry                                                               |
| Defrost Timer Verification and Calibration             | Quarterly                                       | Certified Refrigeration Contractor                      | FRZ-E1, FRZ-E2, FRZ-E3                                          | PM completion certificate; CCMS entry                                                               |
| Data Logger Calibration (NIST-Traceable)               | Annual (January)                                | Accredited Calibration Laboratory                       | SN-LOG-4421 through SN-LOG-4436                                 | NIST-traceable calibration certificate; certificate retained in CCMS and facility Cold Chain binder |
| Full Preventive Maintenance Inspection (Comprehensive) | Annual                                          | Certified Refrigeration Contractor                      | REF-A1 through A5, REF-B1 through B4, FRZ-E1 through E3, PRD-F1 | Full PM inspection report; signed by contractor and DC Supervisor; retained in CCMS                 |
| Alarm System Functional Test                           | Monthly (technician); verified during annual PM | Refrigeration Technician; Certified Contractor (annual) | All assets                                                      | Form 2210-CC (monthly checks); PM report (annual)                                                   |

The following table provides the preventive maintenance schedule for all cold storage assets for the current annual PM cycle (2026–2027):

| Asset ID | Zone / Name                    | Monthly PM Due        | Annual PM Due | Logger Cal. Due | PM Responsible Contractor        |
|----------|--------------------------------|-----------------------|---------------|-----------------|----------------------------------|
| REF-A1   | Cooler-A1 (Meat/Beef)          | 1st Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| REF-A2   | Cooler-A2 (Pork)               | 1st Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| REF-A3   | Cooler-A3 (Poultry)            | 1st Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| REF-A4   | Cooler-A4 (Seafood)            | 1st Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| REF-A5   | Cooler-A5 (Deli)               | 1st Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| REF-B1   | Cooler-B1 (Milk)               | 2nd Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| REF-B2   | Cooler-B2 (Cheese)             | 2nd Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| REF-B3   | Cooler-B3 (Butter/Cream)       | 2nd Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| REF-B4   | Cooler-B4 (Eggs)               | 2nd Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| FRZ-E1   | Freezer-E1 (Seafood/Frozen)    | 3rd Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| FRZ-E2   | Freezer-E2 (Frozen Meals)      | 3rd Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| FRZ-E3   | Freezer-E3 (Frozen Vegetables) | 3rd Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |
| PRD-F1   | Produce-F1                     | 4th Monday each month | Jan 2027      | Jan 2027        | ArcticPro Refrigeration Services |



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## 13. HACCP Integration

The DeCA cold chain handling operation integrates HACCP (Hazard Analysis and Critical Control Points) principles in accordance with USDA FSIS regulatory requirements (9 CFR Part 417) and FDA FSMA preventive control requirements. Three Critical Control Points (CCPs) have been identified and established for cold chain operations. These CCPs represent the points in the cold chain process where control measures are most critical to preventing food safety hazards from reaching commissary patrons.

The HACCP plan for cold chain operations is maintained by the Food Safety Officer, reviewed quarterly, and updated whenever process changes, new product categories, or regulatory changes warrant revision.

### CCP-1: Receiving Temperature Check

| Element                     | Detail                                                                                                                                                      |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Addressed</b>     | Biological: Bacterial growth (Salmonella, E. coli O157:H7, Listeria monocytogenes) resulting from temperature abuse during inbound transport                |
| <b>Critical Limit</b>       | Product temperature at receipt must not exceed the limits defined in Section 11 (e.g., ≤0°F frozen; ≤40°F meat/poultry/seafood/dairy; ≤45°F produce)        |
| <b>Monitoring Frequency</b> | 100% of inbound temperature-sensitive shipments; every pallet or representative sample per DeCA SOP-01 sampling protocol                                    |
| <b>Monitoring Method</b>    | Calibrated probe thermometer (NIST-traceable); transit data logger download review                                                                          |
| <b>Responsible Party</b>    | Receiving Inspector                                                                                                                                         |
| <b>Corrective Action</b>    | Reject non-conforming product; initiate NCR; notify DC Supervisor; notify Food Safety Officer if temp exceeds rejection threshold; document in WMS and CCMS |

| Element                      | Detail                                                                                                                                         |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Verification Activity</b> | DC Supervisor reviews all NCRs weekly; Food Safety Officer audits CCP-1 monitoring records quarterly; thermometer calibration verified monthly |
| <b>Records</b>               | Receiving Temperature Log (DeCA Form 2201-RT); NCR (if applicable); data logger download file; CCMS entry                                      |

## CCP-2: Storage Temperature Monitoring

| Element                      | Detail                                                                                                                                                                                                                |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard Addressed</b>      | Biological: Pathogen growth due to temperature excursion in cold storage; physical: product quality degradation and economic loss                                                                                     |
| <b>Critical Limit</b>        | Cold storage asset temperatures must remain within the acceptable ranges defined in Section 6 at all times. No excursion shall persist beyond 30 minutes without escalation to DC Supervisor and Food Safety Officer. |
| <b>Monitoring Frequency</b>  | Continuous (real-time via automated data loggers and Cold Chain Dashboard); manual verification 3x daily per Section 8                                                                                                |
| <b>Monitoring Method</b>     | Wireless data loggers (SN-LOG-4421 through SN-LOG-4433); Cold Chain Dashboard automated alarms; DeCA Form 2210-CC manual checks                                                                                       |
| <b>Responsible Party</b>     | Refrigeration Technician (TECH-08, TECH-12); Cold Chain Monitor                                                                                                                                                       |
| <b>Corrective Action</b>     | Excursion response per Section 9; product hold in WMS; MKT analysis; Food Safety Officer disposition decision; root cause analysis; corrective action implementation                                                  |
| <b>Verification Activity</b> | Daily CCMS temperature compliance summary reviewed by DC Supervisor; weekly trend analysis by Food Safety Officer; quarterly HACCP review; annual third-party audit                                                   |

| Element | Detail                                                                                                                                 |
|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| Records | DeCA Form 2210-CC; CCMS daily check entries; excursion reports; corrective action records; data logger files; calibration certificates |

### CCP-3: Transport Pre-Cool Verification

| Element               | Detail                                                                                                                                                                                                                                                       |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hazard Addressed      | Biological: Pathogen growth resulting from loading temperature-sensitive product into an inadequately cooled reefer unit; the initial loading temperature sets the baseline for the entire transit cold chain                                                |
| Critical Limit        | Reefer unit interior temperature must be $\leq 35^{\circ}\text{F}$ (perishable) or $\leq -5^{\circ}\text{F}$ (frozen) before loading commences; no exceptions without Food Safety Officer written authorization                                              |
| Monitoring Frequency  | 100% of outbound refrigerated loads; verified before every loading operation                                                                                                                                                                                 |
| Monitoring Method     | Carrier pre-cool documentation review; independent DC dock temperature verification using calibrated thermometer; DeCA Form 2215-LA Loading Authorization Log                                                                                                |
| Responsible Party     | DC Dock Supervisor; Transportation Coordinator                                                                                                                                                                                                               |
| Corrective Action     | Refuse loading until pre-cool temperature is achieved; if carrier cannot achieve pre-cool within 30 minutes, reschedule load; document non-compliance on DeCA Form 2215-LA and report to Transportation Coordinator for carrier corrective action per SOP-04 |
| Verification Activity | Transportation Coordinator reviews Loading Authorization Logs weekly; carrier compliance scorecard reviewed monthly; Food Safety Officer audits CCP-3 records quarterly                                                                                      |
| Records               | DeCA Form 2215-LA; carrier pre-cool                                                                                                                                                                                                                          |

| Element | Detail                                                                       |
|---------|------------------------------------------------------------------------------|
|         | documentation; CCMS entry; carrier corrective action records (if applicable) |

## 14. Food Safety Escalation

When a temperature excursion results in product that cannot be verified as safe for human consumption, or when a cold chain failure event poses a credible risk of distributing unsafe product to commissary patrons, the following food safety escalation procedure shall be executed without delay. This procedure is not discretionary — all personnel identified in the escalation chain have a mandatory duty to act within the specified timeframes.

### 14.1 Immediate Actions (Within 15 Minutes of Disposition Decision)

- WMS Product Hold:** An immediate, system-wide hold shall be placed in the DeCA Warehouse Management System (WMS) on all potentially affected product. The hold shall block the product from picking, packing, shipping, or any release to the commissary distribution network;
- Quarantine Area:** Physically quarantined product shall be moved to the designated quarantine area within the DC (identified by orange quarantine placards and secured with a physical lock or controlled access). The quarantine area location shall be documented in CCMS;
- Chain of Custody:** A chain of custody log shall be initiated in CCMS, documenting every individual who accesses or handles the quarantined product, the date and time of access, and the purpose. No product shall be removed from quarantine without written authorization from the Food Safety Officer.

## **14.2 Notification Requirements (Within 1 Hour of Disposition Decision)**

- The **Food Safety Officer** shall be notified by the DC Supervisor by direct phone call (not email alone) within 1 hour of the disposition decision that product cannot be released;
- The **DC Director** shall be notified by the Food Safety Officer within 1 hour of the disposition decision;
- The **DeCA HQ Supply Chain Operations Division** shall be notified by the DC Director within 2 hours if the affected product quantity or the nature of the event warrants escalation to agency leadership;
- If any potentially affected product has already been shipped to commissary stores prior to the excursion being identified, the Food Safety Officer shall notify the DC Director and DeCA HQ immediately, and initiate a recall/withdrawal assessment without delay.

## **14.3 FDA / USDA FSIS Notification**

The Food Safety Officer shall evaluate the incident against the notification requirements of FDA FSMA Section 204 (enhanced traceability for high-risk foods) and applicable USDA FSIS regulations within 2 hours of the disposition decision. If regulatory notification is required, the Food Safety Officer shall:

- Notify FDA and/or USDA FSIS through the appropriate reporting channels (FDA MedWatch/Safety Reporting Portal for FDA-regulated products; USDA FSIS District Office for FSIS-regulated products);
- Document the notification in CCMS, including the name and title of the regulatory contact, the date and time of notification, and the information provided;
- Retain all documentation related to the notification for the retention periods specified in Section 15.

## 14.4 Product Destruction and Loss Documentation

All product condemned and destroyed as a result of a cold chain failure shall be documented in both CCMS and the DeCA financial management system. The documentation shall include product description, lot numbers, quantities, estimated retail value, and the disposition method (e.g., rendered, landfilled, incinerated). A witness signature from the DC Supervisor or Food Safety Officer is required for all destruction events. The destruction record is retained per Section 15.

 **EXAMPLE — ESCALATION EXECUTED: EXC-2026-07-03 (TRK-002 / LN-005, 14 Jul 2026)**

On 14 July 2026, the excursion event on TRK-002 (Lane LN-005) resulted in a full frozen load quarantine and destruction action. Food Safety Officer was notified by phone at 14:32 local time, within 1 hour of the disposition decision (13:45). DC Director was notified at 14:45. MKT analysis confirmed the 60-minute excursion at +8°F on frozen product constituted an unacceptable cumulative thermal exposure. Entire frozen load was condemned. Chain of custody log initiated; product destroyed under dual-witness supervision. USDA FSIS District Office notified at 15:20. All records retained in CCMS under Event Reference EXC-2026-07-03.

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## 15. Records and Retention

All records generated in connection with cold chain operations, monitoring, excursion response, maintenance, calibration, and food safety actions are official DeCA records and shall be retained in accordance with the timeframes specified in this section. Records may be maintained in electronic form within CCMS (primary repository) with paper backups filed in the facility Cold Chain binder. Electronic records shall be backed up to the DeCA secure records management system no less than weekly.

| <b>Record Type</b>                                    | <b>Primary Form / System</b>           | <b>Retention Period</b>                                         | <b>Governing Authority</b>           |
|-------------------------------------------------------|----------------------------------------|-----------------------------------------------------------------|--------------------------------------|
| Daily Cold Chain Checklists                           | DeCA Form 2210-CC / CCMS               | 2 years minimum                                                 | FDA FSMA (21 CFR Part 1, Subpart O)  |
| Temperature Excursion Reports                         | CCMS Excursion Event Record            | 2 years minimum                                                 | FDA FSMA (21 CFR Part 1, Subpart O)  |
| Root Cause Analysis Records                           | CCMS RCA Record                        | 2 years minimum                                                 | FDA FSMA; DeCA Directive 40-13       |
| Corrective Action Records                             | CCMS Corrective Action Record          | 2 years minimum                                                 | FDA FSMA; DeCA Directive 40-13       |
| Data Logger Downloads (all assets)                    | CCMS Electronic File Archive           | 2 years minimum                                                 | FDA FSMA (21 CFR Part 1, Subpart O)  |
| Data Logger Calibration Certificates                  | CCMS / Facility Cold Chain Binder      | 2 years minimum; or the life of the logger + 2 years            | FDA FSMA; NSF/ANSI 7                 |
| Receiving Temperature Logs / NCRs                     | DeCA Form 2201-RT / WMS / CCMS         | 2 years minimum                                                 | FDA FSMA; USDA FSIS (9 CFR Part 417) |
| Loading Authorization Logs (Pre-Cool Verification)    | DeCA Form 2215-LA / CCMS               | 2 years minimum                                                 | FDA FSMA (21 CFR Part 1, Subpart O)  |
| HACCP Plan and CCP Monitoring Records                 | CCMS / Food Safety Officer Files       | 2 years minimum; HACCP plan retained for life of plan + 2 years | USDA FSIS (9 CFR Part 417); FDA FSMA |
| Preventive Maintenance Records                        | CCMS Maintenance Work Orders           | 2 years minimum                                                 | DeCA Directive 40-13; NSF/ANSI 7     |
| Product Destruction / Loss Records (Financial Impact) | CCMS; DeCA Financial Management System | 3 years minimum                                                 | FAR 4.703 (financial impact records) |
| FDA / USDA Regulatory Notification Records            | CCMS; Food Safety Officer Files        | 3 years minimum                                                 | FDA FSMA Section 204; FAR 4.703      |
| Training Records (Cold Chain Personnel)               | DeCA Training Management System        | Duration of employment + 2 years                                | DeCA Directive 40-13; FDA FSMA       |

### ① RECORDS ACCESS NOTE

All cold chain records are subject to inspection by FDA, USDA FSIS, the DoD Inspector General, and authorized DeCA internal auditors upon request. Records shall be made available within 24 hours of a regulatory inspection request. The DC Supervisor is responsible for ensuring records are organized, complete, and accessible at all times.

## 16. Key Performance Indicators (KPIs)

The following Key Performance Indicators (KPIs) are established to measure the effectiveness of cold chain handling operations across all DeCA distribution centers. KPIs shall be calculated monthly from CCMS data by the Supply Chain Operations Division and reported to the Director, Logistics & Distribution. Facilities failing to meet KPI targets for two or more consecutive months shall be subject to a formal corrective action review.

| KPI                                | Definition                                                                                                                                                  | Target  | Measurement Frequency | Data Source              | Owner                                    |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------------|--------------------------|------------------------------------------|
| <b>Temperature Compliance Rate</b> | Percentage of total data logger readings across all cold storage assets that fall within the acceptable temperature range for the assigned asset during the | ≥ 99.5% | Monthly               | CCMS Data Logger Records | DC Supervisor / Refrigeration Technician |



| KPI                                       | Definition                                                                                                                                                                                       | Target                       | Measurement Frequency | Data Source                     | Owner                                    |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------|---------------------------------|------------------------------------------|
|                                           | reporting period                                                                                                                                                                                 |                              |                       |                                 |                                          |
| <b>Excursion Response Time</b>            | Average time (in minutes) from alarm activation in CCMS/Cold Chain Dashboard to physical technician response at the affected asset, measured across all excursion events in the reporting period | ≤ 15 minutes                 | Monthly               | CCMS Excursion Event Records    | Refrigeration Technician / DC Supervisor |
| <b>Excursion Rate per Asset per Month</b> | Average number of confirmed temperature excursion events per cold storage asset per month across the facility. Excludes planned defrost cycles that are within asset design parameters.          | ≤ 0.5 events / asset / month | Monthly               | CCMS Excursion Event Records    | DC Supervisor / Food Safety Officer      |
| <b>Product Loss Due to Cold Chain</b>     | Total dollar value of product condemned                                                                                                                                                          | \$0 (zero product loss)      | Monthly               | CCMS; DeCA Financial Management | Food Safety Officer / DC Director        |

| KPI                                           | Definition                                                                                                                                                                                                | Target | Measurement Frequency | Data Source                  | Owner                                              |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------|------------------------------|----------------------------------------------------|
| <b>Failure</b>                                | , destroyed, or written off as a direct result of a cold chain temperature excursion or cold chain system failure in the reporting period                                                                 |        |                       | nt System                    |                                                    |
| <b>Daily Check Completion Rate</b>            | Percentage of required daily cold chain checks (Form 2210-CC, 3x daily across all assets) that are completed within the required 30-minute window of the scheduled check time during the reporting period | 100%   | Monthly               | CCMS Daily Check Entries     | Refrigeration Technician / DC Supervisor           |
| <b>Preventive Maintenance Compliance Rate</b> | Percentage of scheduled monthly and annual PM tasks completed on time (within the scheduled                                                                                                               | 100%   | Monthly               | CCMS Maintenance Work Orders | DC Supervisor / Certified Refrigeration Contractor |

| KPI                                          | Definition                                                                                                                                                                | Target | Measurement Frequency | Data Source                               | Owner                                    |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------|-------------------------------------------|------------------------------------------|
|                                              | PM week) across all cold storage assets during the reporting period                                                                                                       |        |                       |                                           |                                          |
| <b>Carrier Pre-Cool Compliance Rate</b>      | Percentage of inbound reefer loads for which the carrier achieved the required pre-cool temperature prior to dock arrival, as documented on DeCA Form 2215-LA             | ≥ 99%  | Monthly               | DeCA Form 2215-LA / CCMS                  | Transportation Coordinator               |
| <b>Data Logger Calibration Currency Rate</b> | Percentage of deployed data loggers (SN-LOG-4421 through SN-LOG-4436) with a current, valid NIST-traceable calibration certificate (calibrated within the past 12 months) | 100%   | Monthly               | CCMS Asset Register / Calibration Records | Refrigeration Technician / DC Supervisor |

## 17. Training Requirements

All personnel with cold chain handling responsibilities, as defined in Section 5 of this SOP, must complete the training requirements specified in this section prior to performing cold chain duties independently. Training records shall be maintained in the DeCA Training Management System and are subject to audit by the Food Safety Officer and external regulatory inspectors. Failure to maintain required training currency is a compliance deficiency requiring immediate corrective action by the DC Supervisor.

### 17.1 Required Training by Role

| Training Requirement                                                                                                                                                                         | Applicable Roles                                                                                      | Frequency                                                                             | Certification / Completion Standard                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>HACCP Certification</b> — Accredited HACCP training program covering hazard analysis, CCP identification, critical limits, monitoring, corrective action, verification, and recordkeeping | Food Safety Officer, DC Supervisor, Refrigeration Technician, Cold Chain Monitor, Receiving Inspector | Initial certification required before independent duty; recertification every 3 years | Certificate from NEHA, NRFSP, or equivalent accredited HACCP program; copy retained in Training Management System        |
| <b>Cold Chain SOP Training (DeCA-SC-SOP-07)</b> — Comprehensive review of this SOP, including temperature standards, excursion response, data logger operation, and HACCP CCPs               | All personnel listed in Section 5                                                                     | Initial (before independent duty); annual refresher; upon any SOP revision            | Signed DeCA Training Acknowledgment Form; scored $\geq$ 80% on SOP-07 knowledge assessment administered by DC Supervisor |
| <b>Refrigeration Safety and Equipment Training</b> — Safe operation of refrigeration                                                                                                         | Refrigeration Technician, Cold Chain Monitor, Picker/Packer                                           | Initial; annual refresher                                                             | Facility-administered training; attendance record in Training Management                                                 |

| <b>Training Requirement</b>                                                                                                                                                                                                            | <b>Applicable Roles</b>                                                                                                        | <b>Frequency</b>                                                       | <b>Certification / Completion Standard</b>                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| equipment, data logger operation and download procedures, alarm response, and cold storage GMP                                                                                                                                         |                                                                                                                                |                                                                        | System                                                                                                                                                                        |
| <b>Temperature Excursion Response Drill</b> — Simulated excursion event requiring technician to demonstrate correct alarm response, product hold placement, CCMS documentation, and supervisor notification within required timeframes | Refrigeration Technician, Cold Chain Monitor, DC Supervisor                                                                    | Quarterly (4x per year)                                                | Drill conducted and evaluated by DC Supervisor or Food Safety Officer; pass/fail result documented in Training Management System; failed drills require immediate remediation |
| <b>New Hire Cold Chain Orientation / OJT Rotation</b> — 30-day on-the-job training rotation covering all cold chain assets, CCMS operation, data logger procedures, daily check execution, and excursion response                      | All new hires assigned to cold chain duties (Refrigeration Technician, Cold Chain Monitor, Receiving Inspector, Picker/Packer) | Once (upon hire); must be completed before independent duty assignment | Completion sign-off by supervising Refrigeration Technician (e.g., TECH-08) and DC Supervisor; OJT checklist retained in Training Management System                           |
| <b>FDA FSMA Sanitary Transportation Rule Training</b>                                                                                                                                                                                  | Transportation Coordinator, Receiving Inspector, DC Supervisor, Food Safety Officer                                            | Initial; refresher when regulatory updates are issued                  | FDA FSMA online training module completion certificate; retained in Training Management System                                                                                |
| <b>Carrier Cold</b>                                                                                                                                                                                                                    | All carrier drivers                                                                                                            | Upon carrier                                                           | Carrier driver sign-                                                                                                                                                          |

| Training Requirement               | Applicable Roles        | Frequency                    | Certification / Completion Standard                                                                     |
|------------------------------------|-------------------------|------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Chain Requirements Briefing</b> | operating on DeCA lanes | onboarding; annual refresher | off on DeCA Carrier Cold Chain Requirements Acknowledgment Form; retained by Transportation Coordinator |

## 17.2 Training Non-Compliance

Any personnel found to be performing cold chain duties without current required training certifications shall be immediately removed from cold chain duties by the DC Supervisor until training is completed. The DC Supervisor shall document the non-compliance in the employee's training record and implement a corrective plan to bring the employee into training compliance within 30 calendar days. Recurring training non-compliance shall be escalated to the Director, Logistics & Distribution.

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## 18. Revision History

| Revision | Date           | Description of Change                                                                                                                                                                                      | Author / Prepared By             | Approved By                        |
|----------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------------|
| 1.0      | 01 August 2026 | Initial release of DeCA-SC-SOP-07 Cold Chain Handling. Document establishes standardized cold chain handling, monitoring, excursion response, transportation, HACCP integration, maintenance, records, and | Supply Chain Operations Division | Director, Logistics & Distribution |

| Revision | Date | Description of Change                                                                                                                                                                                                                                                                                                             | Author / Prepared By | Approved By |
|----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------|
|          |      | training requirements for all DeCA distribution centers and refrigerated transportation lanes. Incorporates FDA FSMA Sanitary Transportation Rule (21 CFR Part 1, Subpart O), USDA FSIS regulations, DeCA Directive 40-13, and applicable NSF/ASHRAE standards. Supersedes all prior informal facility-level cold chain guidance. |                      |             |

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## Approval and Certification

The signatures below certify that this Standard Operating Procedure has been reviewed for accuracy, completeness, and regulatory compliance, and is hereby approved for implementation effective 01 August 2026.

**Director, Logistics & Distribution**

Defense Commissary Agency Date:

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**Facility Food Safety Officer** Defense  
Commissary Agency Date:

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**Chief, Supply Chain Operations**

**Division** Defense Commissary Agency Date:

\_\_\_\_\_

**Distribution Center Director** Defense  
Commissary Agency Date:

\_\_\_\_\_

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**Document Control:** DeCA-SC-SOP-07 | Revision 1.0 | Effective: 01 August 2026 |  
Next Review: 01 August 2027

**Classification:** UNCLASSIFIED // FOR OFFICIAL USE ONLY (FOUO) — Handle in accordance with DoD Instruction 5400.11.

**Controlling Office:** Supply Chain Operations Division, Defense Commissary Agency, Fort Gregg-Adams, VA — For questions or proposed revisions, contact the Supply Chain Operations Division Cold Chain Program Manager.

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